UNIVERSITETET I OSLO

IN5340 / IN9340 Lecture 3
Parameter Estimation

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Contents

What do we learn

- Parameter estimation
- ch 6 in Therrien and the whole book of Kay.

S. M. Kay.
Fundamentals of Statistical Signal Processing: Estimation Theory, Volume I.
Prentice Hall International Editions, 1993.

C. W. Therrien.
Discrete Random Signals and Statistical Signal Processing.
Prentice Hall, 1992.

Additional information

- The course `EE 522` given by Prof. Mark Fowler at Binghamton University has excellent lecture notes
- “Google” for EE522 and download / read:
  - `Notes #1a Probability Review`
  - `Notes #2 Introduction to Estimation`
  - `Notes #3 Ch 2 MVUE`
  - `Notes #12 Ch 7A MLE`
- The lecture notes on probability gives an excellent review
- The lecture notes on estimation covers more or less everything
- See also *Estimation theory* in Wikipedia

`https://ws.binghamton.edu/fowler/fowler%20personal%20page/EE522.htm`

Motivation

Estimation theory

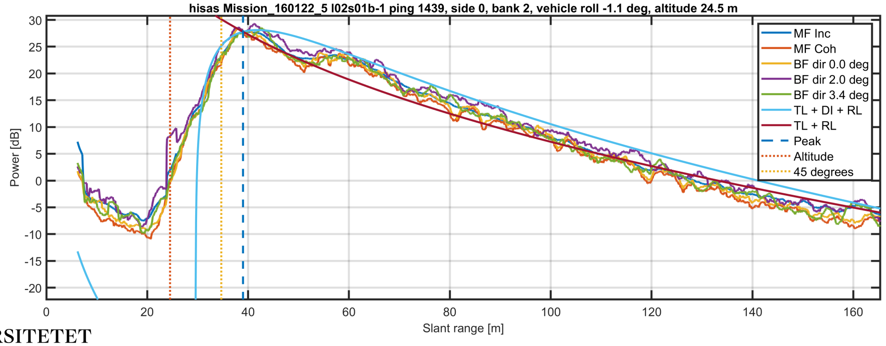
- A branch of statistics / statistical signal processing that deals with estimating the values of parameters based on measured/empirical data that has a random component.
- The parameters describe an underlying physical setting in such a way that their value affects the distribution of the measured data.
- An estimator attempts to approximate the unknown parameters using the measurements.

Example

- Example: Sonar backscattering from the seafloor
- The task: Associate physical quantities to observations

Example

- How to estimate the signal level and shape
- How to fit a model to the data the best possible way
- How to calculate the accuracy of the estimate



Example applications of estimation theory

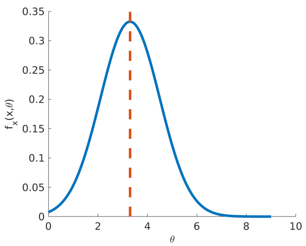
- **Radar:** estimating the position of a target (an aeroplane)
- **Sonar:** estimating the direction of arrival from a noise source (a propeller on a submarine or the voice from a marine mammal)
- **Speech recognition:** estimate the value of each phoneme (sound unit)
- **Image analysis:** estimate the position and orientation of an object (in robotics / computer vision)
- **Biomedicine:** estimate the heart rate
- **Wireless communications:** estimate the carrier frequency of a signal for demodulation
- **Seismics:** estimate the distance to an oil deposit within the layers

Contents

Mathematical description

- Assume a set of observations organised in a vector
  $$\mathbf{x} = [x_1, x_2, x_3, \ldots, x_N]^T$$
  which depends on an unknown parameter θ
- This can either be an ensemble of a random variable, a timeseries of a random process etc
- We wish to determine θ based on the data
  $$\hat{\theta} = g(\mathbf{x})$$
  where g is some function.
- This is the problem of *parameter estimation*
- Note that $\hat{\theta}$ is a random variable

Mathematical description



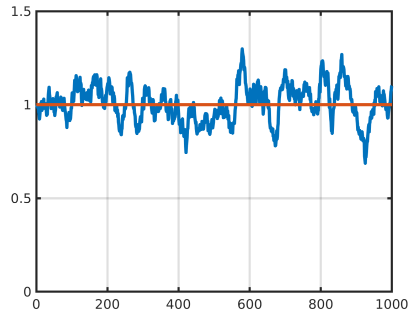
- The data is inherently random
- Described by the probability density function
- The PDF is dependent of the unknown parameter θ
  $$f_x(\mathbf{x}; \theta)$$
- An N-dimensional PDF parameterised by θ
- Since $\hat{\theta}$ is a random variable, we describe it with a PDF (or rather, consider the PDF a function of the parameter)

Probability Density Function

The N-dimensional PDF parameterised by θ describes "everything".
In general not given - choose a PDF that captures the essense of reality.

Example

- Assume a signal in additive noise
  $$x[n] = s[n, \theta] + w[n]$$
  where $s[n, \theta]$ is the signal as a function of some parameter θ, and $w[n]$ is additive noise
- Simple example: The signal is a constant in noise
  $$x[n] = A + w[n]$$
- Challenge: How to estimate the signal A
- A reasonable estimator for A would be
  $$\hat{A} = \frac{1}{N}\sum_{n=1}^{N} x[n]$$
- How good is the estimator?



Mathematical approach to solve the example

- The estimated value produced by an estimator is a random variable
- The performance of the estimator can therefore only be completely described statistically by its PDF
- Assuming that the noise $w[n]$ is white zero-mean Gaussian noise, the probability density function for *a single sample* becomes

$$f_x(x[1], A) = \frac{1}{\sqrt{2\pi\sigma_x^2}} \exp\left(-\frac{(x[1]-A)^2}{2\sigma_x^2}\right)$$

  where σ_x^2 is the variance of the noise process $w[n]$

Additive White Gaussian Noise

- *Additive White Gaussian Noise* (AWGN)
  - Additive ⇔ is added to the signal
  - White ⇔ flat power spectral density
  - Gaussian ⇔ Gaussian probability density function
  - Often specified zero-mean (without loss of generality)
- Why Gaussian? Each observation is often based on a large number of random events
- The central limit theorem states that summing independent random variables will cause the PDF to approache the Normal distribution
- Recall the example with additive noise in the sonar. Although man-made, with a non-flat PSD, the PDF was still Gaussian.

Mathematical approach to solve the example

- Back to the example with N samples: A multidimensional PDF
- Since the noise is AWGN, each sample is statistically independent, and the joint PDF becomes separable

$$\begin{aligned} f_x(\mathbf{x}, A) &= f_x(x[1], A) f_x(x[2], A) \ldots f_x(x[N], A) \\ &= \prod_{n=1}^{N} \frac{1}{\sqrt{2\pi\sigma_x^2}} \exp\left(-\frac{(x[n]-A)^2}{2\sigma_x^2}\right) \\ &= \frac{1}{(2\pi\sigma_x^2)^{N/2}} \exp\left(-\frac{1}{2\sigma_x^2}\sum_{n=1}^{N}(x[n]-A)^2\right) \end{aligned}$$

  (Rule: A product becomes a sum in the exponent)

Mathematical approach to solve the example

- Our simple example: The signal is a constant in noise

  $x[n] = A + w[n]$
- How good is the sample mean as an estimator?
- The estimated value is a random variable.
- Desire: The expected value of the estimator becomes the true value

  $E\{\hat{\theta}\} = \theta$
- Since we have the PDF, this is possible to calculate

$$E\{\hat{A}\} = E\left\{\frac{1}{N}\sum_{n=1}^{N} x[n]\right\} = \frac{1}{N}\sum_{n=1}^{N} \underbrace{E\{x[n]\}}_{=A} = A$$

- Hence, the sample mean has an expected value of the true value

Terminology

Probability theory

- Assume a probability density function model
- Describe how the random process will most likely behave
- Estimate characteristics (mean, variance etc) from PDF
- Characterise estimator performance based on the assumed model
- No data

Statistics

- Given a set of data (a random sequence)
- Calculate how the data did behave
- Calculate Sample mean, variance etc
- Estimate fitted values, trends etc
- No model (PDF)

Properties of estimators

The performance of an estimator can be characterised by certain properties

- Bias
- Variance
- Consistency
- Efficiency
- Mean square error
- Confidence interval

These properties are important to consider when choosing which estimator, and which optimisation criteria for estimators

Bias of estimator

- An estimator is said to be *unbiased* if
  $$E\{\hat{\theta}\} = \theta \qquad a < \theta < b$$
  where (a, b) denotes the interval of possible values of θ
- Otherwise, the estimator is *biased* with bias
  $$b(\theta) = E\{\hat{\theta}\} - \theta$$
- An estimator is said to be *asymptotically unbiased* if
  $$\lim_{N\to\infty} E\{\hat{\theta}_N\} = \theta$$
  the suspect notation with subscript N indicates that the estimate is a function of the number of observations
- Unbiased estimators tends to have symmetric PDFs around the true value

When do we have bias?

- A typical case when bias occurs, is when the signal model or the noise model is incorrect
- Example: Consider again a constant unknown value A in AWGN
  $$x[n] = A + w[n]$$
- This time, let the noise have a non-zero mean component
- We assume zero-mean noise and use the sample mean estimator
  $$\hat{A} = \frac{1}{N}\sum_{n=1}^{N} x[n]$$
- It is then obvious that the estimator becomes biased (does not approach the correct value) simply because our chosen model does not represent the data properly

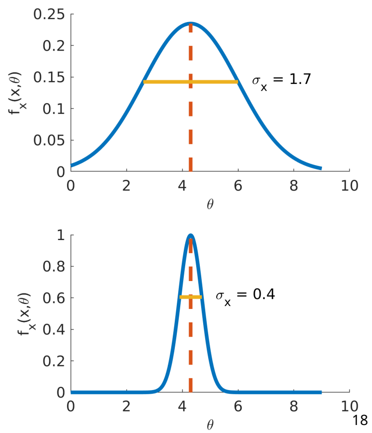
Variance of estimator

Remember (again) that the estimate itself is a random variable

- The *variance* of an estimator is another important property

$$\mathrm{Var}(\hat{\theta}) = E\{|\hat{\theta} - E\{\hat{\theta}\}|^2\}$$

- If an estimate has low variance then there is high probability that the estimate is close to the true value
- An estimator is said to be *efficient* with respect to another estimator, if it has a lower variance
- Note: $2\sigma_x$ is not the Full Width Half Maximum (FWHM) for the normal distribution: $FWHM = 2\sqrt{2\ln(2)}\sigma_x \approx 2.35\sigma_x$



Variance of estimator - example

- A simple example: a constant unknown value A in zero-mean AWGN

$$x[n] = A + w[n]$$

- The variance becomes

$$\mathrm{Var}(\hat{A}) = \mathrm{Var}\left(\frac{1}{N}\sum_{n=1}^{N} x[n]\right) = \frac{1}{N^2}\sum_{n=1}^{N}\mathrm{Var}(x[n]) = \frac{1}{N^2}\sum_{n=1}^{N}\sigma_x^2 = \frac{\sigma_x^2}{N}$$

- Remember that if the RVs x and y are uncorrelated, then

$$\mathrm{Var}(x+y) = \sigma_x^2 + \sigma_y^2 = \mathrm{Var}(x) + \mathrm{Var}(y)$$

- Since the estimate is unbiased, and the variance decreases with increasing N, the estimate is *consistent*

Mean square error

- The *mean square error* of an estimator is given by

$$\mathrm{MSE}(\hat{\theta}) = E\left\{(\hat{\theta} - \theta)^2\right\}$$

- The bias of the estimator is given as

$$b(\theta) = E\{\hat{\theta}\} - \theta$$

- Simple calculus gives us

$$\begin{aligned}
\mathrm{MSE}(\hat{\theta}) &= E\left\{\left[(\hat{\theta} - E\{\hat{\theta}\}) + \underbrace{(E\{\hat{\theta}\} - \theta)}_{b(\theta)}\right]^2\right\} \\
&= E\left\{\left[\hat{\theta} - E\{\hat{\theta}\}\right]^2\right\} + 2b(\theta)\underbrace{E\left\{\hat{\theta} - E\{\hat{\theta}\}\right\}}_{=0} + b^2(\theta) \\
&= \mathrm{Var}(\hat{\theta}) + b^2(\theta)
\end{aligned}$$

Terminology - again

From the Wikipedia page on *precision and accuracy*

- Precision is a description of random errors, a measure of statistical variability
- Accuracy is a description of systematic errors
- In our language:
  Precision represents variance
  Accuracy represents bias

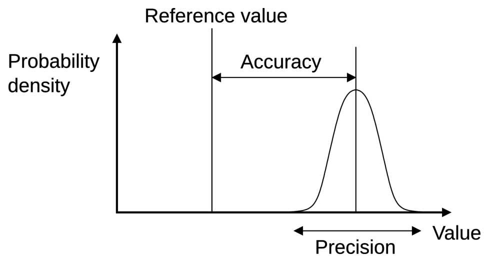


Illustration from Wikipedia

We will continue to use *Bias* and *Variance*

Estimators - Short Summary

Unbiased estimate The expected value of the estimate is the true value

Asymptotically Unbiased Estimate The expected value of the estimate approaches the true value when the number of samples approaches infinity

Consistent Estimate Unbiased estimate where the variance decreases with increasing number of samples

Efficiency An estimate is said to be *efficient* with respect to another estimate if the variance is lower

Example: Estimating the correlation function

- The autocorrelation function for a WSS real random process is
$$R_{xx}(\tau) = E\{x(t)x(t+\tau)\}$$
- Assume a finite length random sequence of length N
- An estimator for R_{xx} is
$$\hat{R}_{xx}[l] = \frac{1}{N-|l|}\sum_{n=0}^{N-1-|l|} x[n+|l|]x[n] \qquad |l| < N$$
- What is the performance of this estimator?
- See the function `xcorr` in Matlab

Estimator properties for the correlation function

- The expectation of the estimator of the autocorrelation function (the sample autocorrelation sequence) is
$$\begin{aligned} E\{\hat{R}_{xx}[l]\} &= \frac{1}{N-|l|}\sum_{n=0}^{N-1-|l|} E\{x[n+|l|]x[n]\} \\ &= \frac{1}{N-|l|}\sum_{n=0}^{N-1-|l|} R_{xx}[l] = R_{xx}[l] \end{aligned}$$
- Hence, the estimator is *unbiased*
- See the parameters `biased` and `unbiased` for the function `xcorr` in Matlab

Estimator properties for the correlation function

- The variance of the estimator is slightly more tricky to calculate, since fourth order moments appears
$$\begin{aligned} \mathrm{Var}(\hat{R}_{xx}[l]) &= E\{|\hat{R}_{xx}[l] - E\{\hat{R}_{xx}[l]\}|^2\} \\ &= E\{\hat{R}_{xx}^2[l] - 2\hat{R}_{xx}[l]E\{\hat{R}_{xx}[l]\} + E\{\hat{R}_{xx}[l]\}^2\} \\ &= E\{\hat{R}_{xx}^2[l] - R_{xx}^2[l]\} = E\{\hat{R}_{xx}^2[l]\} - R_{xx}^2[l] \end{aligned}$$
remembering that $E\{\hat{R}_{xx}[l]\} = R_{xx}[l]$
- Now,
$$\begin{aligned} E\{\hat{R}_{xx}^2[l]\} &= E\left\{\frac{1}{(N-|l|)^2}\sum_{n=0}^{N-1-|l|} x[n+|l|]x[n]\sum_{m=0}^{N-1-|l|} x[m+|l|]x[m]\right\} \\ &= \frac{1}{(N-|l|)^2}\sum_{n=0}^{N-1-|l|}\sum_{m=0}^{N-1-|l|} E\{x[n+|l|]x[n]x[m+|l|]x[m]\} \end{aligned}$$

Estimator properties for the correlation function

- The Isserlis theorem (from 1918) provides a simplification for Gaussian zero mean random processes

$$E\{X_1X_2X_3X_4\} = E\{X_1X_2\}E\{X_3X_4\} + E\{X_1X_3\}E\{X_2X_4\} + E\{X_1X_4\}E\{X_2X_3\}$$

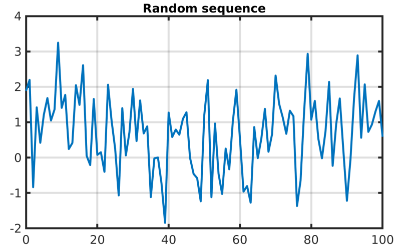
- Hence,

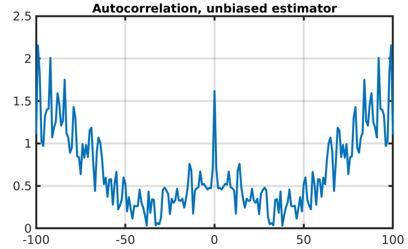
$$\mathrm{Var}(\hat{R}_{xx}[l]) \sim E\{\hat{R}^2_{xx}[l]\} \sim \frac{1}{(N-|l|)^2}R^2_{xx}[l]$$

- The variance of the estimator is “well behaved” since it decreases with increasing N
- The variance increases however with increasing l

Estimator properties for the correlation function

- The variance of the estimator turns out to be “well behaved”
- The variance decreases with increasing number of samples, and the estimator is unbiased: The estimator is *consistent*
- There is, however, a fundamental problem with the estimator
- The property $|R_{xx}(\tau)| \leq R_{xx}(0)$ does not always hold





Example: The correlation function - new attempt

- Another estimator for R_{xx} is

$$\hat{R}_{xx}[l] = \frac{1}{N}\sum_{n=0}^{N-1-|l|} x[n+|l|]x[n] \qquad |l| < N$$

- The expectation of this estimator is

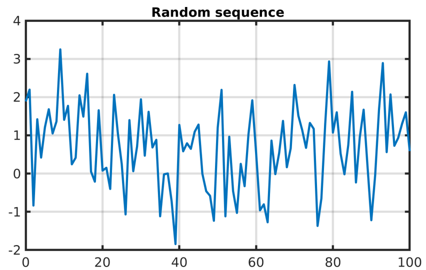
$$E\{\hat{R}_{xx}[l]\} = \frac{1}{N}\sum_{n=0}^{N-1-|l|} E\{x[n+|l|]x[n]\} = \frac{N-|l|}{N}R_{xx}[l]$$

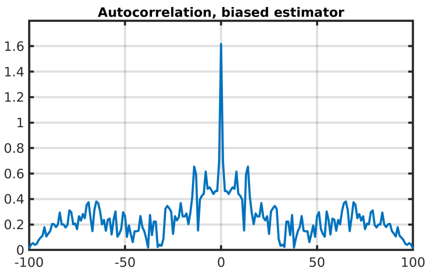
- Hence, the estimator is *biased*
- This estimator is however, *asymptotically unbiased*

$$\lim_{N\to\infty} E\{\hat{R}_{xx}[l]\} = \lim_{N\to\infty} \frac{N-|l|}{N}R_{xx}[l] = R_{xx}[l]$$

Estimator properties for the correlation function

- The variance of the estimator is again “well behaved”
- Now, the autocorrelation sequence is maximum at zero lag. Always.





Contents

Optimal estimators

Parameter estimation grand goal

Finding an optimal estimator and assessing its performance

- There are a number of different optimisation criteria
- The optimal estimator does not always exsist...
- We will look into a few different types of estimators
  - The Minimum Variance Unbiased Estimator (MVUE)
  - The Maximum Likelihood Estimator (MLE)
  - The Least Squares Estimator (LSE) - suboptimum

Classes of estimation

Classical methods

- If we view the parameter to estimate θ as deterministic (non-random), it is referred to as *classical estimation*
- This gives us no ability to include a priori information about θ
- Examples:
  - minimum variance
  - maximum likelihood
  - least squares techniques

Bayesian methods

- If we view θ as random, it is referred to as *Bayesian estimation*
- Allows use of a priori information about θ
- Examples:
  - minimum mean squared error
  - maximum a posteriori estimator
  - Wiener filter
  - Kalman filter

Bayesian methods

When do we actually need a Bayesian approach

- When we wish to add a priori knowledge about the desired estimate θ
- Assume θ to be a random variable
- Assign a given *prior* PDF to the estimate
- Since we actually add information, the estimator may become better than obtainable in classical estimation where the estimate is assumed to be deterministic and unknown.

Will be touched upon later in the course:

- Detection: Bayes detection
- Optimal and adaptive filters: Wiener filter, MMSE

Minimum Variance Unbiased Estimation (MVUE)

MVUE Estimator design criteria

Of all unbiased estimators, find the one that has minimum variance

- Unbiased estimator

  $$E\{\hat{\theta}\} = \theta \qquad \text{for all } \theta$$

- Variance

  $$\mathrm{Var}(\hat{\theta}) \quad \text{as small as possible}$$

- As we remember, the mean square error is

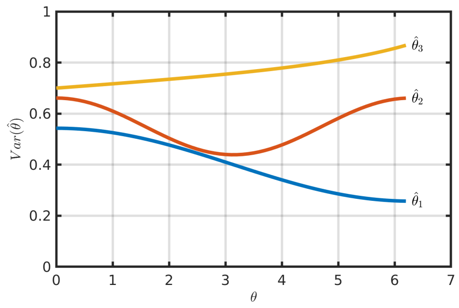
  $$\mathrm{MSE}(\hat{\theta}) = \mathrm{Var}(\hat{\theta}) + b^2(\theta)$$

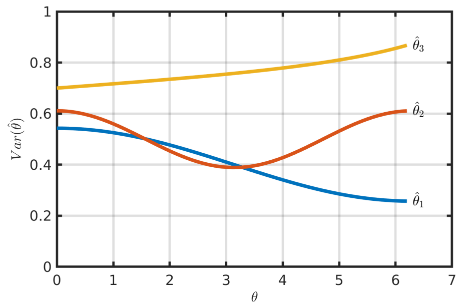
- MVU could also be called minimum MSE Unbiased estimator

Existence of Minimum Variance Unbiased Estimators

Sometimes there are no Minimum Variance Unbiased Estimators

- Because there are no unbiased estimators
- Because none of the unbiased estimators have minimum variance over all parameter values





Finding the Minimum Variance Unbiased Estimator

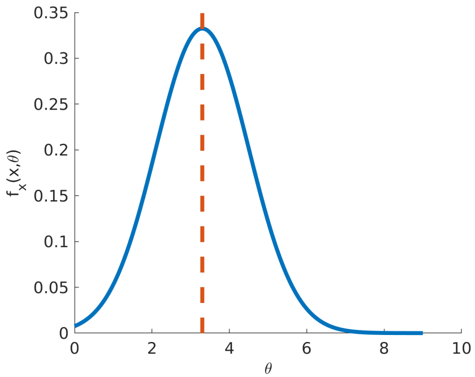
- Even if the MVU estimator exists, there are no simple “algorithm” to find it.
- There are a few approaches:
  - Determine the theoretical limit (lower bound) and check if an estimator satisfies it
  - Restrict the class of estimators to be linear. Then find the MVU estimator
  - ++
- See chapters 2 and 5 in Kay for more details on MVUE
- We will continue to the next class of estimators

Maximum Likelihood Estimation

The density function $f_x(\mathbf{x};\theta)$, when viewed as a function of θ for fixed values of observations, is called the *likelihood function*

- The *maximum likelihood estimate* is simply the parameter that maximises the likelihood function

  $$\hat{\theta}_{ml} = \underset{\theta}{\mathrm{argmax}}\, f_x(\mathbf{x};\theta)$$



Maximum Likelihood Estimation - simple example

- Consider again the simple example of a constant unknown value A in noise
  $$x[n] = A + w[n]$$
  where we have a single observation $N = 1$
- The PDF becomes
  $$f_x(x[1], A) = \frac{1}{\sqrt{2\pi\sigma_x^2}} \exp\left(-\frac{(x[1] - A)^2}{2\sigma_x^2}\right)$$
- The maximum likelihood estimate is simply $\hat{A}_{ml} = x[1]$

Maximum Likelihood Estimation - better example

- Consider the better example of a constant unknown value A in noise
  $$x[n] = A + w[n]$$
  where we have a sequence of N observations
- The PDF becomes
  $$f_x(\mathbf{x}, A) = \frac{1}{(2\pi\sigma_x^2)^{N/2}} \exp\left(-\frac{1}{2\sigma_x^2} \sum_{n=1}^{N} (x[n] - A)^2\right)$$
- We want to maximise this function
- The logarithm is a monotonic function. Maximising the likelihood function is equivalent to maximising the *log likelihood function*

The log likelihood function

- The logarithm of the likelihood function becomes
  $$\ln f_x(\mathbf{x}, A) = -\frac{N}{2} \ln(2\pi\sigma_x^2) - \frac{1}{2\sigma_x^2} \sum_{n=1}^{N} (x[n] - A)^2$$
- This is a quadratic function in A with a single maximum where the derivative is zero
  $$\begin{aligned}\frac{\partial \ln f_x(\mathbf{x}, A)}{\partial A} &= \frac{\partial}{\partial A}\left(-\frac{1}{2\sigma_x^2} \sum_{n=1}^{N} (x[n] - A)^2\right) \\ &= \frac{1}{2\sigma_x^2} \frac{\partial}{\partial A}\left(-\sum_{n=1}^{N} x[n]^2 + \sum_{n=1}^{N} 2Ax[n] - \sum_{n=1}^{N} A^2\right) \\ &= \frac{1}{\sigma_x^2} \sum_{n=1}^{N} x[n] - \frac{A}{\sigma_x^2} N\end{aligned}$$

The maximum likelihood estimate

- By setting the derivative to zero
  $$\frac{\partial \ln f_x(\mathbf{x}, A)}{\partial A} = 0$$
  we then get
  $$\frac{1}{\sigma_x^2} \sum_{n=1}^{N} x[n] - \frac{A}{\sigma_x^2} N = 0$$
  which leads to the *Maximum Likelihood Estimator*
  $$\hat{A}_{ml} = \frac{1}{N} \sum_{n=1}^{N} x[n]$$
- The sample mean is the maximum likelihood estimate

The maximum likelihood estimate stated generally

- Whenever the likelihood function is continuous and the maximum does not occur at a boundary, the maximum likelihood estimate can be obtained by the solution of either equations:

$$\left.\frac{\partial f_x(\mathbf{x},\theta)}{\partial\theta}\right|_{\theta=\hat{\theta}_{ml}} = 0$$

$$\left.\frac{\partial \ln f_x(\mathbf{x},\theta)}{\partial\theta}\right|_{\theta=\hat{\theta}_{ml}} = 0$$

- Straightforward to implement given known PDF

Maximum Likelihood Estimator (MLE) Summary

- By far the most popular approach
- Provides a “turn-the-crank” procedure for developing the algorithm
- Can be implemented for complicated estimation problems
- Asymptotically unbiased
- Asymptotically efficient (approaches the lower bound)
- Approximately the MVU estimator due to its approximate efficiency
- Practically optimal (large data record and/or high SNR)
- Easily extended to multiple parameters

Maximum Likelihood Estimation - final example

- Consider a sinusoid with unknown phase ϕ in additive noise

$$x[n] = A\cos(2\pi f_0 n + \phi) + w[n]$$

with known frequency f_0 and amplitude A

- The MLE is found by maximizing the Likelihood function

$$f_x(\mathbf{x},A) = \frac{1}{(2\pi\sigma_x^2)^{N/2}} \exp\left(-\frac{1}{2\sigma_x^2}\sum_{n=0}^{N-1}(x[n] - A\cos(2\pi f_0 n + \phi))^2\right)$$

- This is equivalent to minimizing

$$J(\phi) = \sum_{n=0}^{N-1}(x[n] - A\cos(2\pi f_0 n + \phi))^2$$

Maximum Likelihood Estimation - final example (2)

- Differentiation with respect to ϕ

$$\frac{\partial J(\phi)}{\partial\phi} = 2\sum_{n=0}^{N-1}(x[n] - A\cos(2\pi f_0 n + \phi))A\sin(2\pi f_0 n + \phi)$$

- Setting equal to zero gives

$$\sum_{n=0}^{N-1} x[n]\sin(2\pi f_0 n + \phi) = A\sum_{n=0}^{N-1}\cos(2\pi f_0 n + \phi)\sin(2\pi f_0 n + \phi)$$

- The right hand side:

$$A\sum_{n=0}^{N-1}\cos(2\pi f_0 n + \phi)\sin(2\pi f_0 n + \phi) \approx 0$$

for f_0 not near 0 or $1/2$ since sin and cos are orthogonal when summed over full cycles

Maximum Likelihood Estimation - final example (3)

- This leaves the following

$$\sum_{n=0}^{N-1} x[n] \sin(2\pi f_0 n + \phi) = 0$$

- Using the *sine of sum* rule $\sin(\alpha + \beta) = \cos\beta \sin\alpha + \sin\beta \cos\alpha$ we obtain a direct expression for the MLE of phase

$$\hat{\phi}_{ml} = -\tan^{-1}\left\{\frac{\sum_{n=0}^{N-1} x[n] \sin(2\pi f_0 n)}{\sum_{n=0}^{N-1} x[n] \cos(2\pi f_0 n)}\right\}$$

- See EE522 slides from Binghamton's University and Kay's book on Estimation for details (Example 7.6 and 7.16)

Contents

Summary of lecture IV

- Parameter estimation
- Additive White Gaussian Noise
- Probability theory vs statistical analysis
- Bias, variance, consistency
- Unbiased, asymptotically unbiased
- Mean square error
- Estimation of the sample autocorrelation
- Optimal estimators
- Classical and Bayesian estimation
- Minimum Variance Unbiased Estimator
- The likelihood function
- Maximum Likelihood Estimator